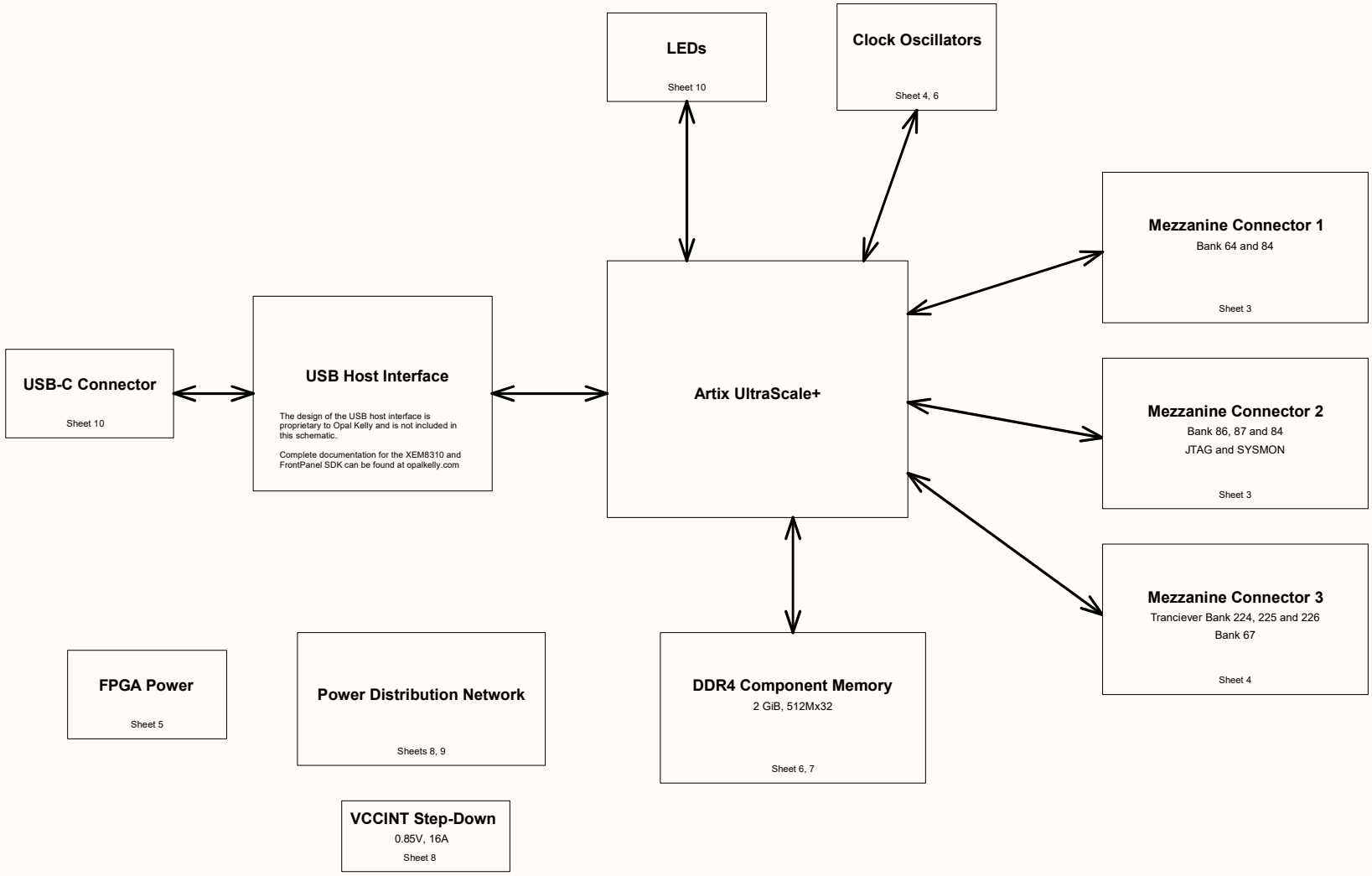
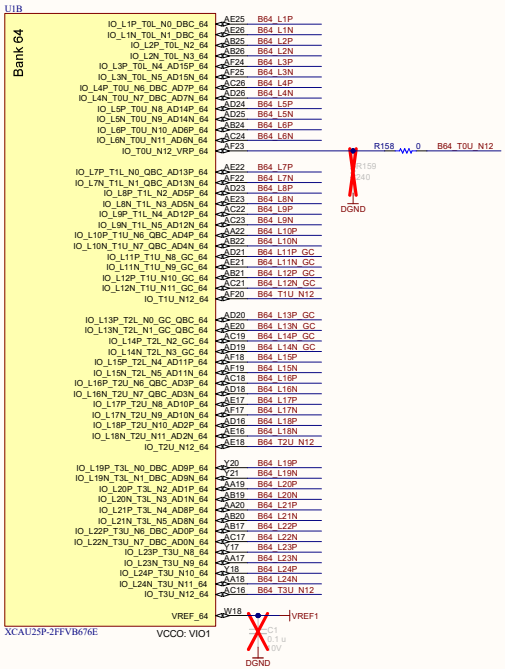


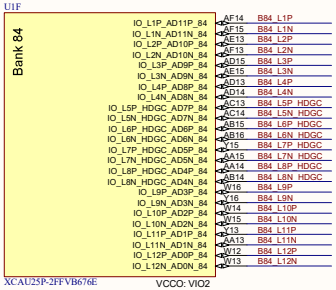
Title: *XEM8310 - Block Diagram*			Opal Kelly 13500 SW 72nd Ave STE 100 Portland, OR 97223 http://www.opalkelly.com
Size: C	Number: 2	Revision: A 03	
Date: 5/17/2022	Time: 7:01:53 AM	Sheet 2 of 12	
Drawn by: RCA			

Opal Kelly

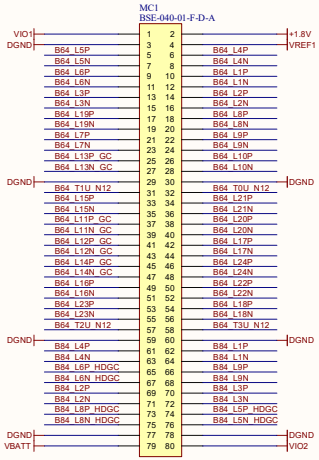
HP Bank 64



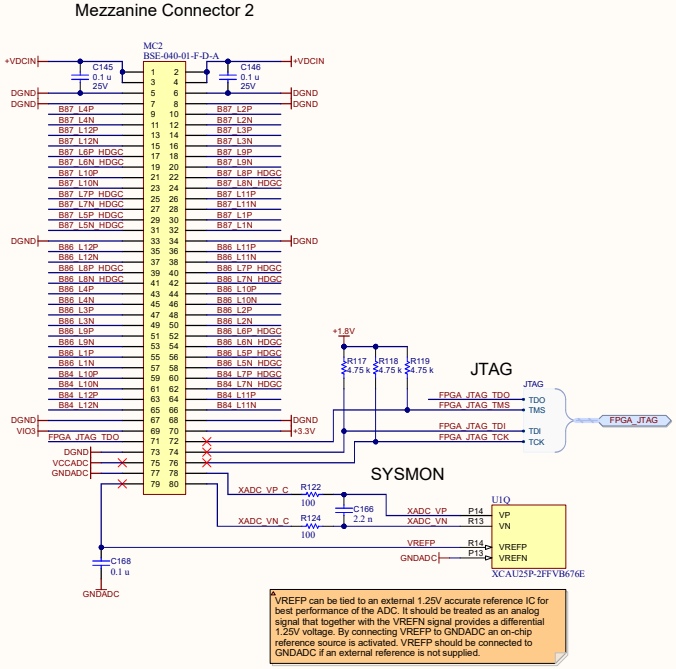
HD Bank 84



Mezzanine Connector 1

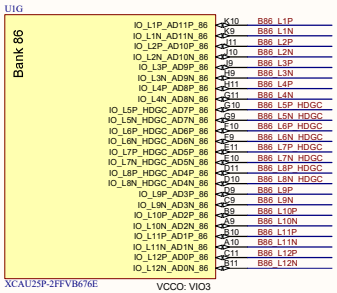


Mezzanine Connector 2

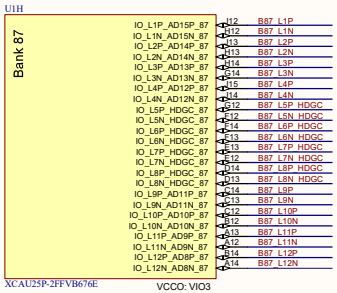


VREFP can be tied to an external 1.25V accurate reference IC for best performance of the ADC. It should be treated as an analog signal that together with the VREFN signal provides a differential 1.25V voltage. By connecting VREFP to GNDADC an on-chip reference source is activated. VREFP should be connected to GNDADC if an external reference is not supplied.

HD Bank 86

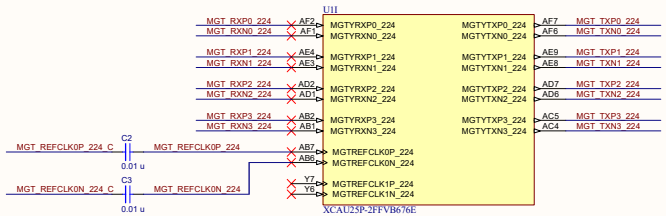


HD Bank 87

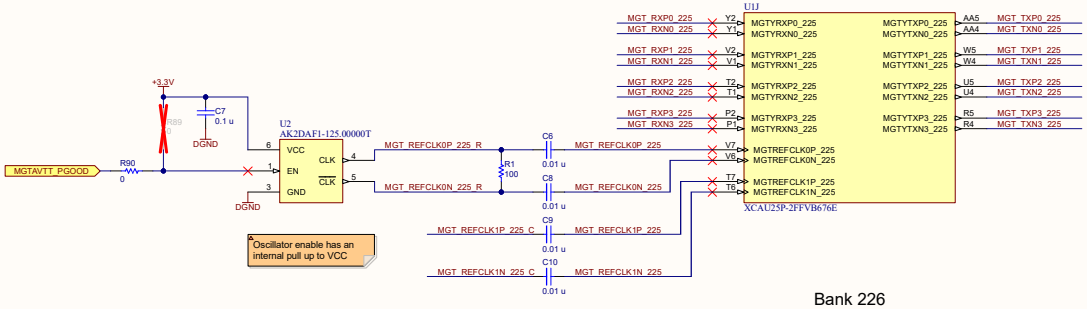


Title: XEM8310 - FPGA Expansion MC			Opal Kelly 13500 SW 72nd Ave STE 100 Portland, OR 97223 http://www.opalkelly.com
Size: C	Number: 3	Revision: A 03	
Date: 5/17/2022	Time: 7:01:53 AM	Sheet 3 of 12	
Drawn by: RCA			

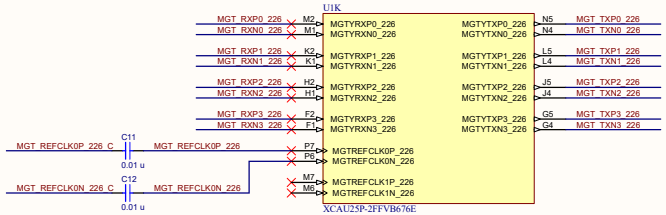
Bank 224

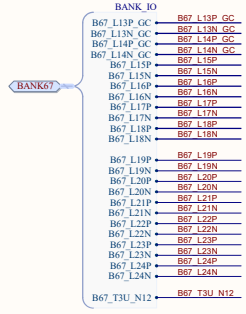


Bank 225



Bank 226

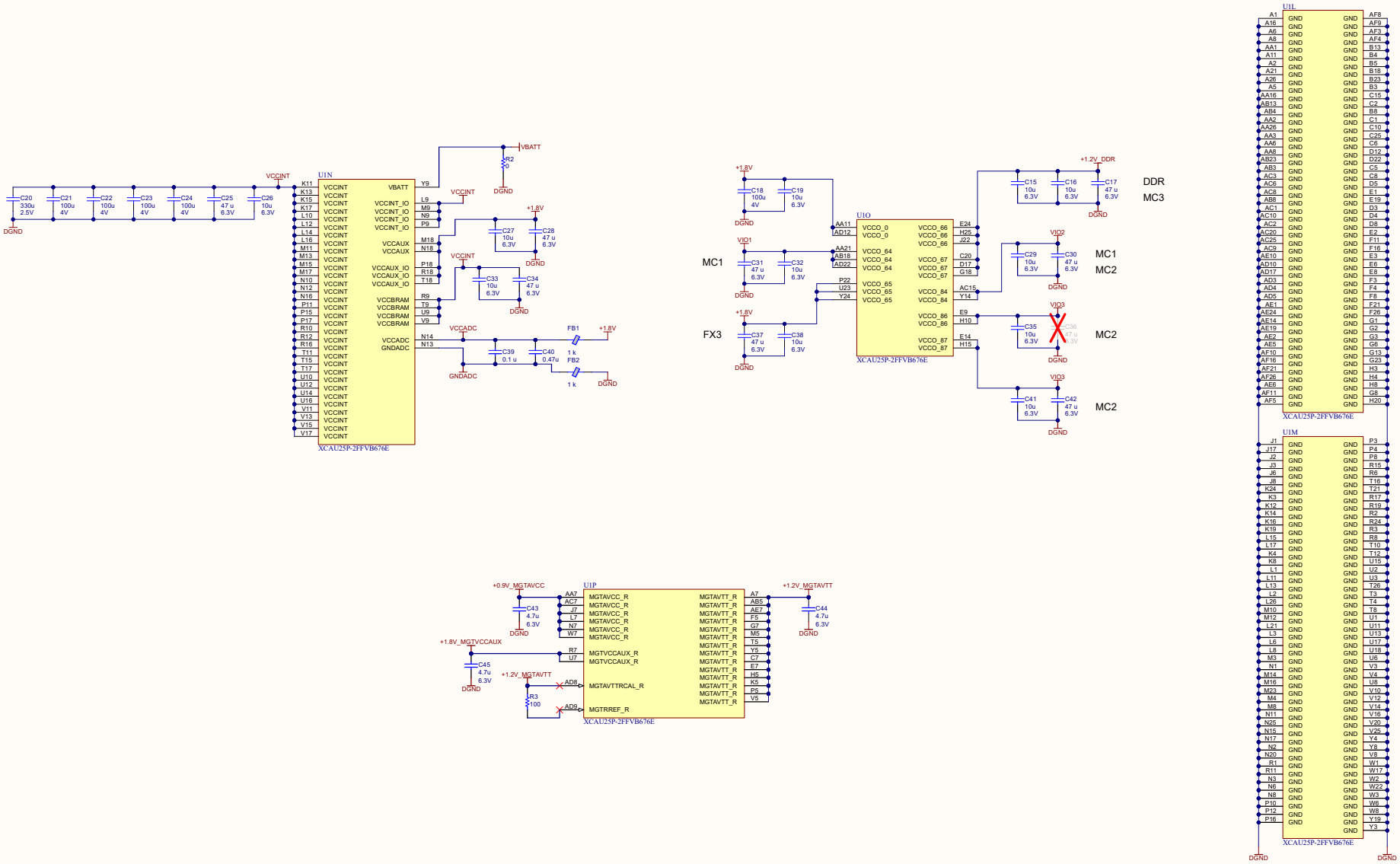




Mezzanine Connector 3



Title: XEM8310 - FPGA Expansion Transceiver			Opal Kelly 13500 SW 72nd Ave STE 100 Portland, OR 97223 http://www.opalkelly.com
Size: C	Number: 4	Revision: A 03	
Date: 5/17/2022 Time: 7:01:53 AM		Sheet 4 of 12	
Drawn by: RCA			



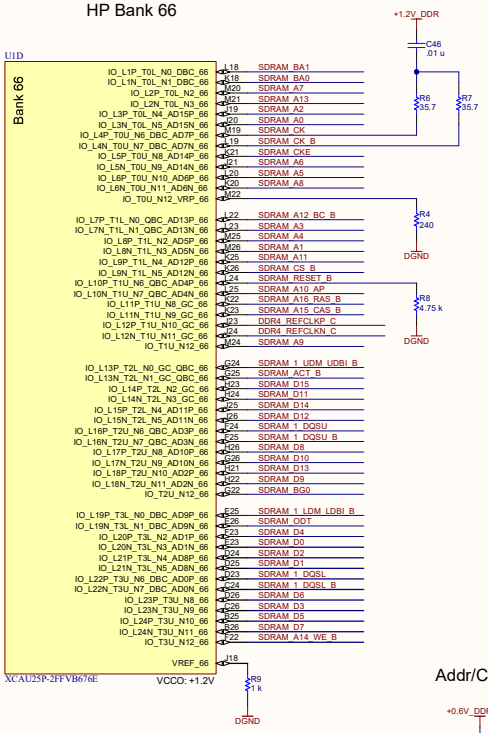
Title: XEM8310 - FPGA Power			Opal Kelly 13500 SW 72nd Ave STE 100 Portland, OR 97223 http://www.opalkelly.com
Size: C	Number: 5	Revision: A 03	
Date: 5/17/2022 Time: 7:01:53 AM	Sheet 5 of 12		
Drawn by: RCA			

Pinout constraints:
- Addr/cntl signals can go on any pins in bank 66 (except VREF)
- CK must be on a P/N pair
- Data byte groups can go on any FPGA bank byte groups (T0:T3) in any order on any bank (66, 67)
- xDM/xDBI must be on pin N0 in the associated byte group
- DQSx clock pair must be on pins N6/N7 in the associated byte group
- Pins N1 and N12 cannot be used for DQ signals
- PAR, TEN, and ALERT_B are not supported on Ultrascale and not connected.

DDR4 Design References:
Xilinx KCU105
Xilinx KCU116
Micron TN-40-40

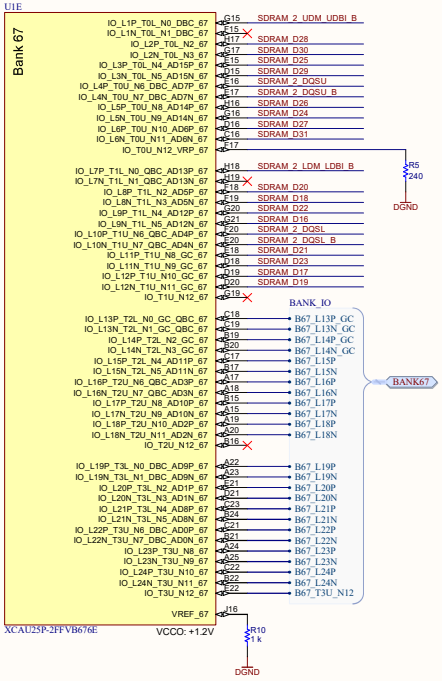
For pinout constraints, see "DDR4 Pin Rules" in PG150.

HP Bank 66

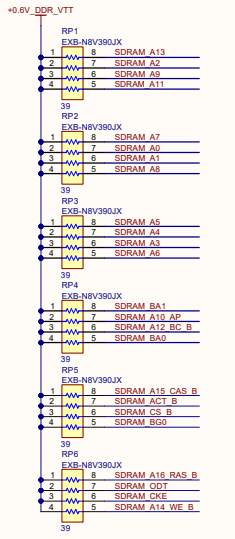


HP Bank 67

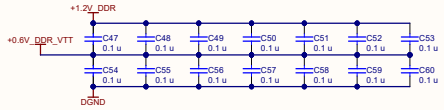
AU10P, AU15P and AU20P board variants dont have BANK 67

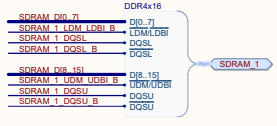


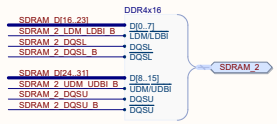
Addr/Ctrl Termination Resistors

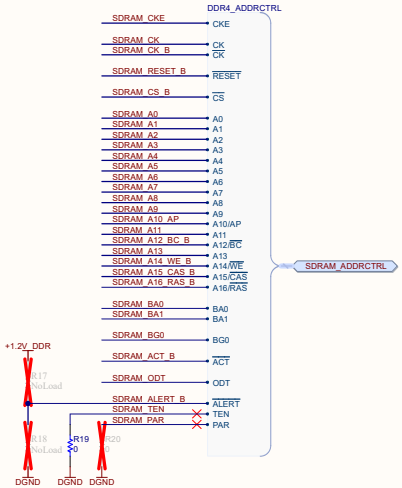


VTT Bypass Capacitors



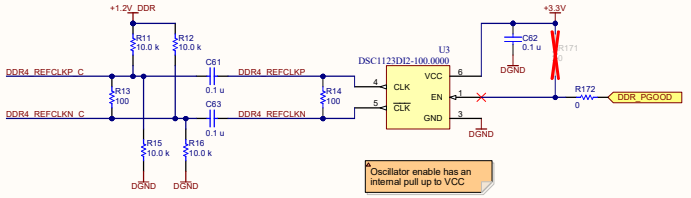






Design Note:
The AC coupling network rebiases the clock signal for use on the 1.2V bank.

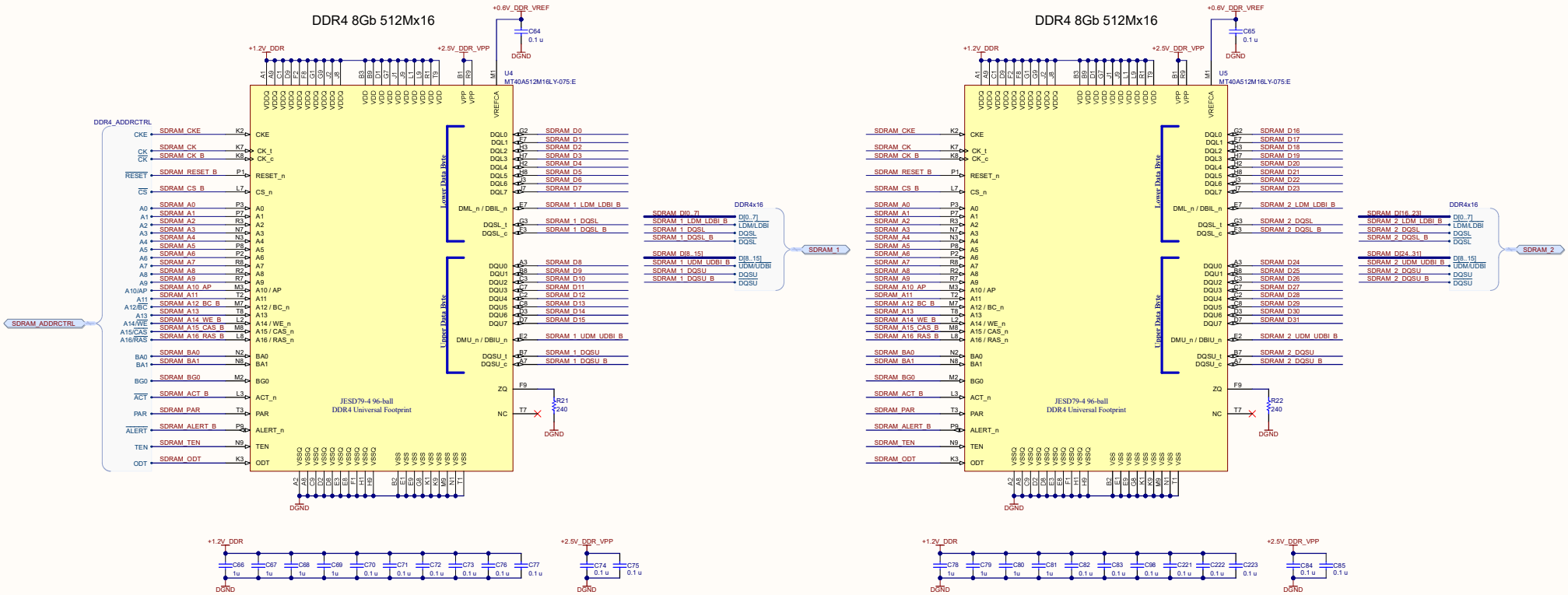
See UG471 v1.6 p.92 (LVDS and LVDS_25 Receiver Termination).



Oscillator enable has an internal pull up to VCC

Title: XEM8310 - FPGA Memory			Opal Kelly 13500 SW 72nd Ave STE 100 Portland, OR 97223 http://www.opalkelly.com
Size: C	Number: 6	Revision: A 03	
Date: 5/17/2022	Time: 7:01:53 AM	Sheet 6 of 12	
Drawn by: RCA			

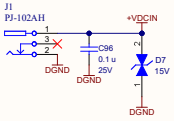
Opal Kelly



Title: XEM8310 - Memory			Opal Kelly 13500 SW 72nd Ave STE 100 Portland, OR 97223 http://www.opalkelly.com
Size: C	Number: 7	Revision: A 03	
Date: 5/17/2022	Time: 7:01:53 AM	Sheet 7 of 12	
Drawn by: RCA			

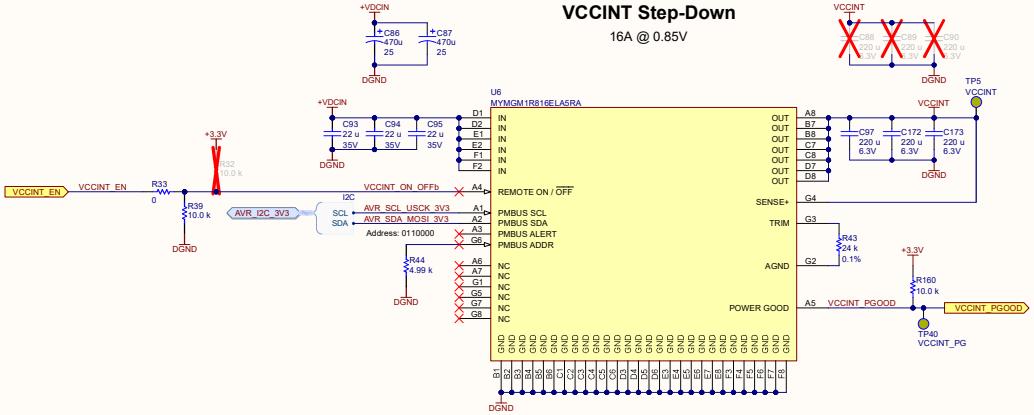
Power Connector

+7.5 to +15 VDC



VCCINT Step-Down

16A @ 0.85V

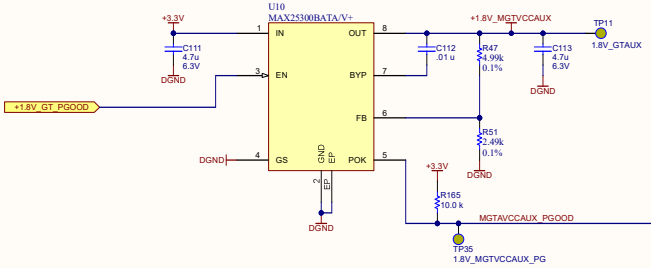


The recommended UltraScale power-on sequence:
VCCINT -> VCCINT_IO/VCCBRAM --> VCCAUX/VCCAUX_IO --> VCCO

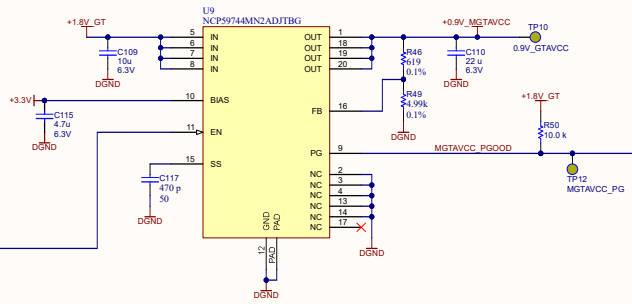
For the GTH or GTY transceivers:
VCCINT/MGTAVCC --> MGTAVTT

Transceiver Regulators

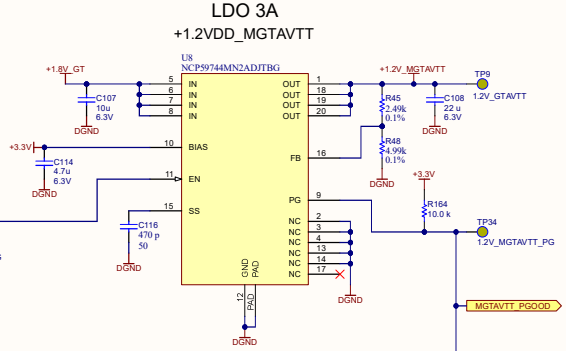
LDO 0.5A
+1.8VDD_MGTVCCAUX



LDO 3A
+0.9VDD_MGTAVCC

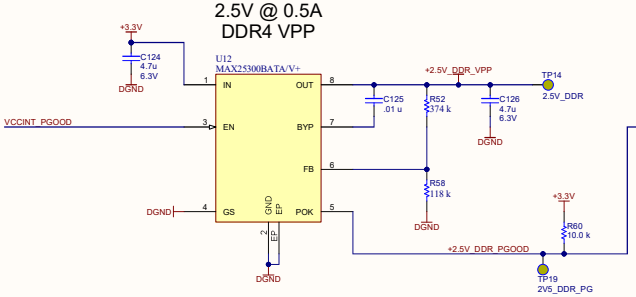


LDO 3A
+1.2VDD_MGTAVTT

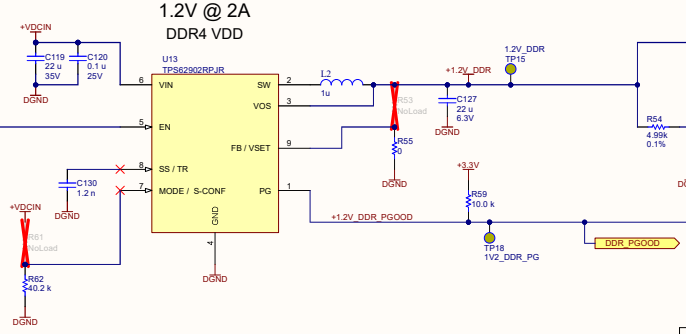


DDR Regulators

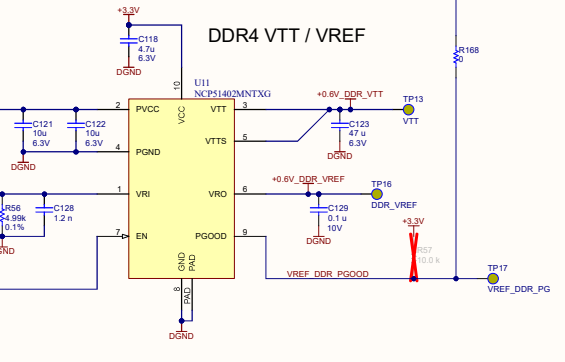
2.5V @ 0.5A
DDR4 VPP



1.2V @ 2A
DDR4 VDD



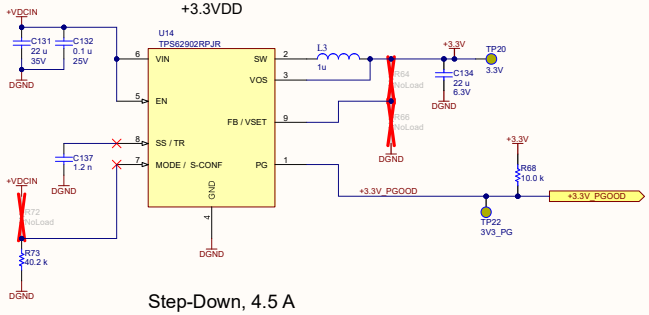
DDR4 VTT / VREF



Title: XEM8310 - Power 1			Opal Kelly 13500 SW 72nd Ave STE 100 Portland, OR 97223 http://www.opalkelly.com
Size: C	Number: 8	Revision: A 03	
Date: 5/17/2022	Time: 7:01:53 AM	Sheet 8 of 12	
Drawn by: RCA			

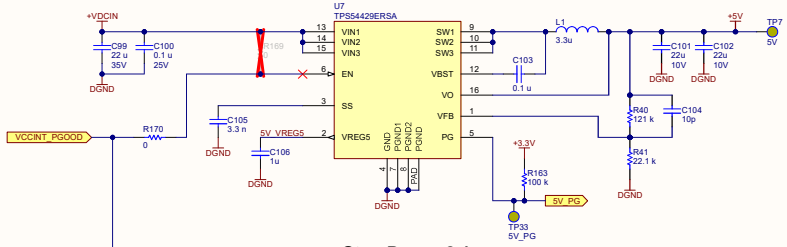
Step-Down, 2 A

+3.3VDD



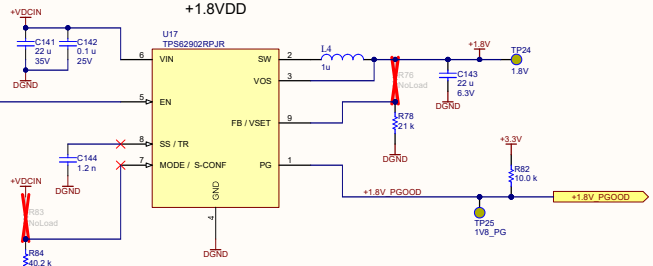
Step-Down, 4.5 A

+5VDD



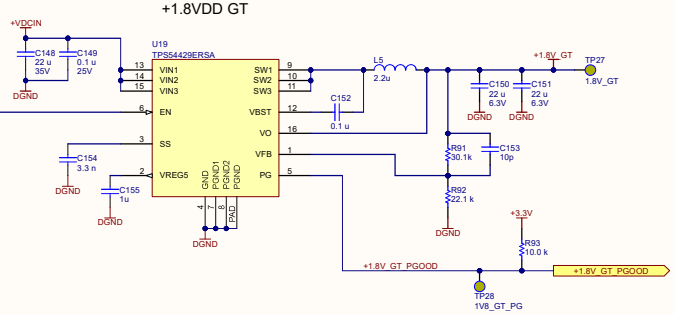
Step-Down, 2 A

+1.8VDD

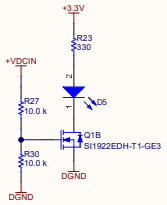


Step-Down, 4.5 A

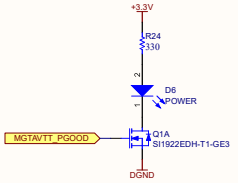
+1.8VDD GT



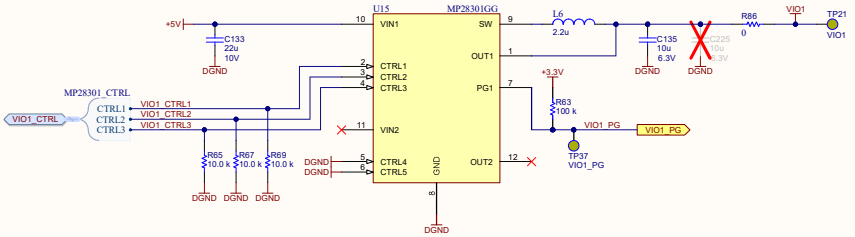
INPUT POWER LED



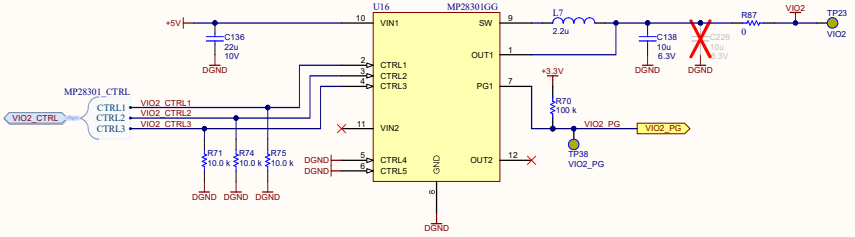
POWER-GOOD LED



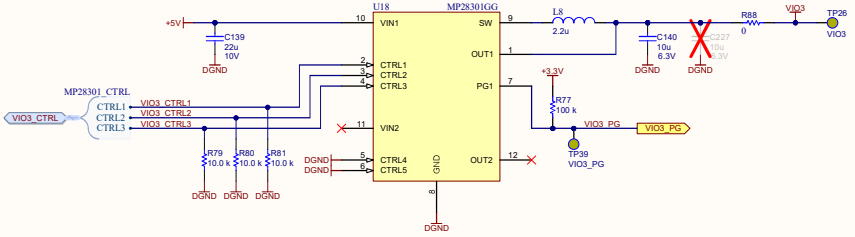
VIO1 Step-Down, 0.7 A



VIO2 Step-Down, 0.7 A

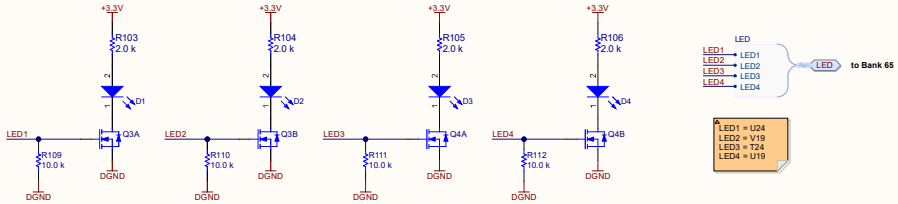


VIO3 Step-Down, 0.7 A

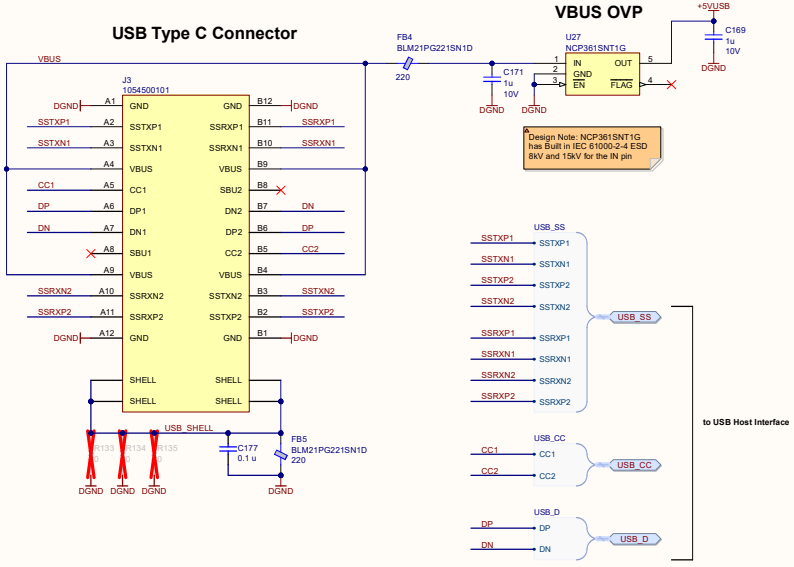


Title: XEM8310 - Power 2			Opal Kelly 13500 SW 72nd Ave STE 100 Portland, OR 97223 http://www.opalkelly.com
Size: C	Number: 9	Revision: A 03	
Date: 5/17/2022 Time: 7:01:54 AM		Sheet 9 of 12	
Drawn by: RCA			

User LEDs



USB Type C Connector



VBUS OVP

Title: XEM8310 - USB / LEDs			Opal Kelly 13500 SW 72nd Ave STE 100 Portland, OR 97223 http://www.opalkelly.com
Size: C	Number: 10	Revision: A 03	
Date: 5/17/2022 Time: 7:01:54 AM		Sheet 10 of 12	
Drawn by: RCA			